After Dark on a Country Road

Three decades of road safety in Australia, 1989–2021

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In 1989, 2,800 people died on Australian roads. In 2021 this was reduced to 843, a 70 per cent drop in just one generation. Tens of thousands of lives have been saved through seatbelts, random breath testing, safer cars and better infrastructure. By any measure this is one of the great public-health achievements of modern Australia.

But the headline doesn't tell the whole story. Not all states did so. Not all weeks were equal with the same improvement. The benefit of the road did not accrue evenly. The six charts in this report highlight that same and point to where the next decade of road safety needs to focus.

> **The big idea.** *The road death narrative is not a national problem anymore that being spread evenly. It is a story of where progress stopped on weekend nights, on country roads, in the seat of a young male driver in a state with a long way to go.*

Key numbers: 52,843 total deaths 1989–2021 | 70% national decline | 12,576 weekend night deaths | 5,019 young male drivers killed

Thirty years, 70 per cent

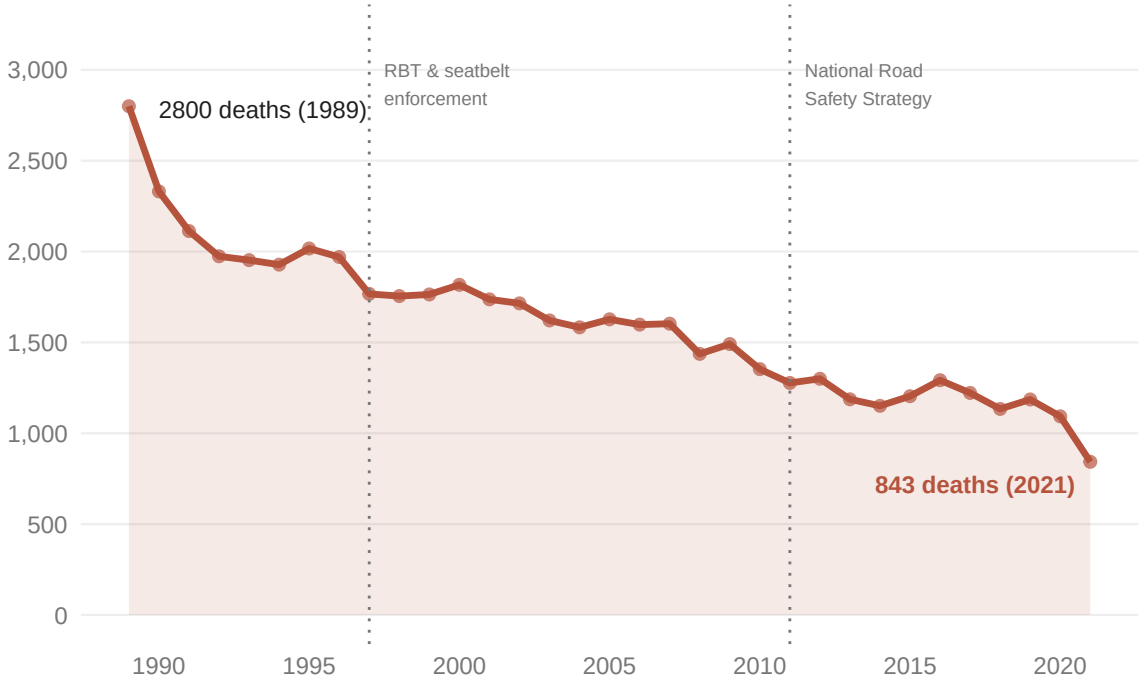


Figure 1: Annual road fatalities in Australia, 1989–2021 (filled line chart).

The fall is indeed substantial. Every important policy step had an impact in the form of a higher segment on the line. However, since around 2010 rates of improvement have declined, suggesting the easy gains have already been made.

The pattern inside the year

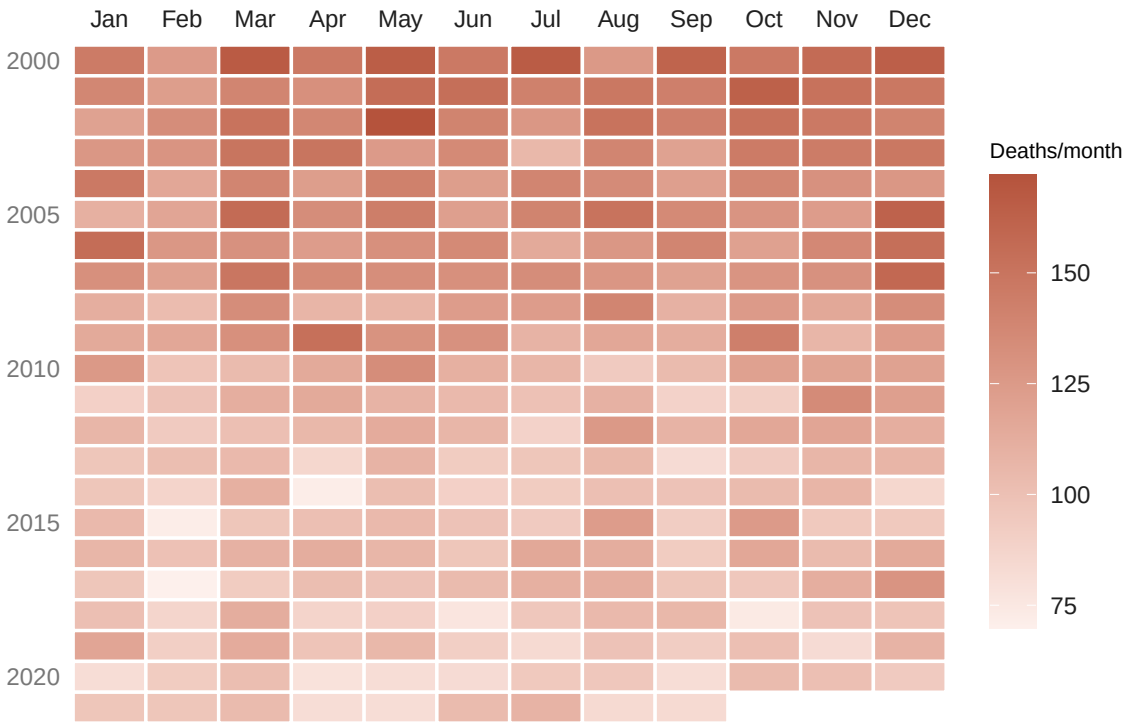


Figure 2: Calendar heatmap — fatalities by year and month, 2000–2021. Darker red means more deaths.

This heatmap is a 22-year capture of heat into one grid. The cells represent a month each. The columns: January, which falls in the Australian summer holiday period, is consistently darker than the winter months. Reading the rows: the 2000–2004 band at the top is deep red; by 2018–2021 almost every cell is pale. The improvement is on system-wide basis, but the peak in January never completely disappeared.

Not all states kept pace

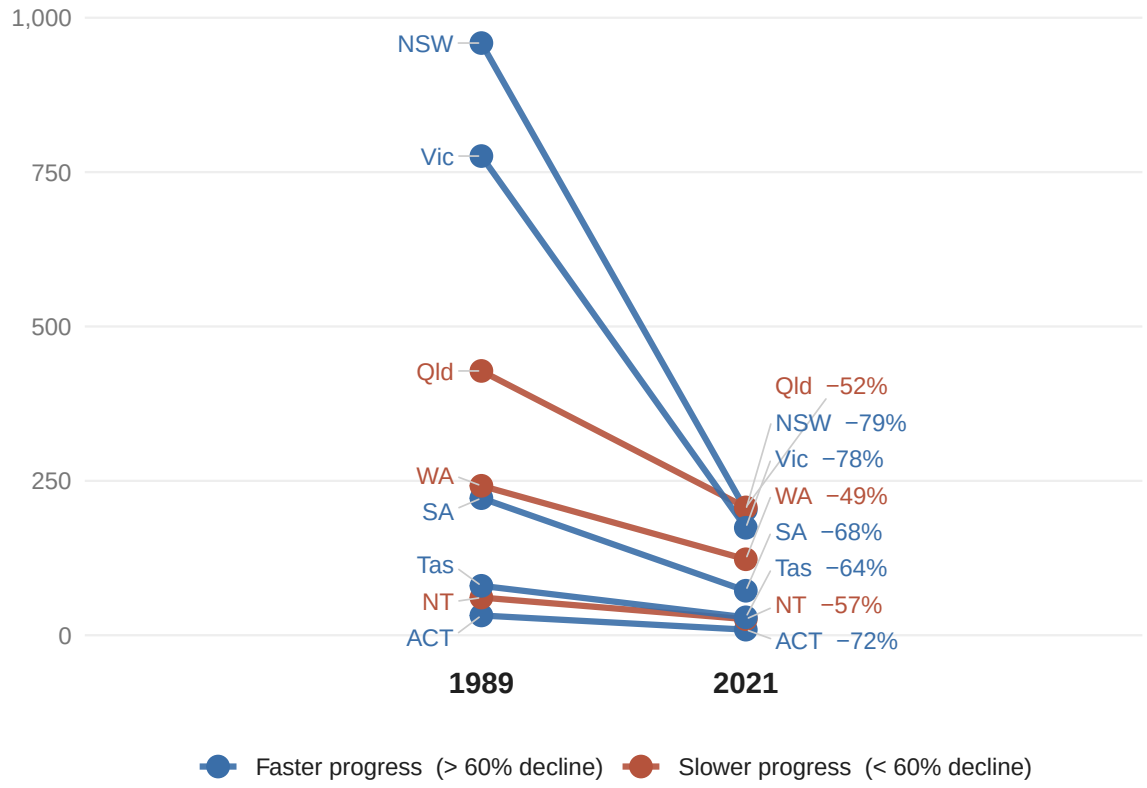


Figure 3: Slope chart — state fatalities in 1989 vs 2021. Percentage change is labelled on the right. Red = less than 60 per cent decline.

NSW and Victoria (blue) reduced both their totals by around 78 per cent. Western Australia's decrease was just 49 per cent and Queensland's was 52 per cent. These are the states where the time it takes to get from town to town is longest, and where the rural roads are least divided. The same geography that made roads dangerous in 1989, has made improvement slower since.

Saturday night, one car, no witnesses

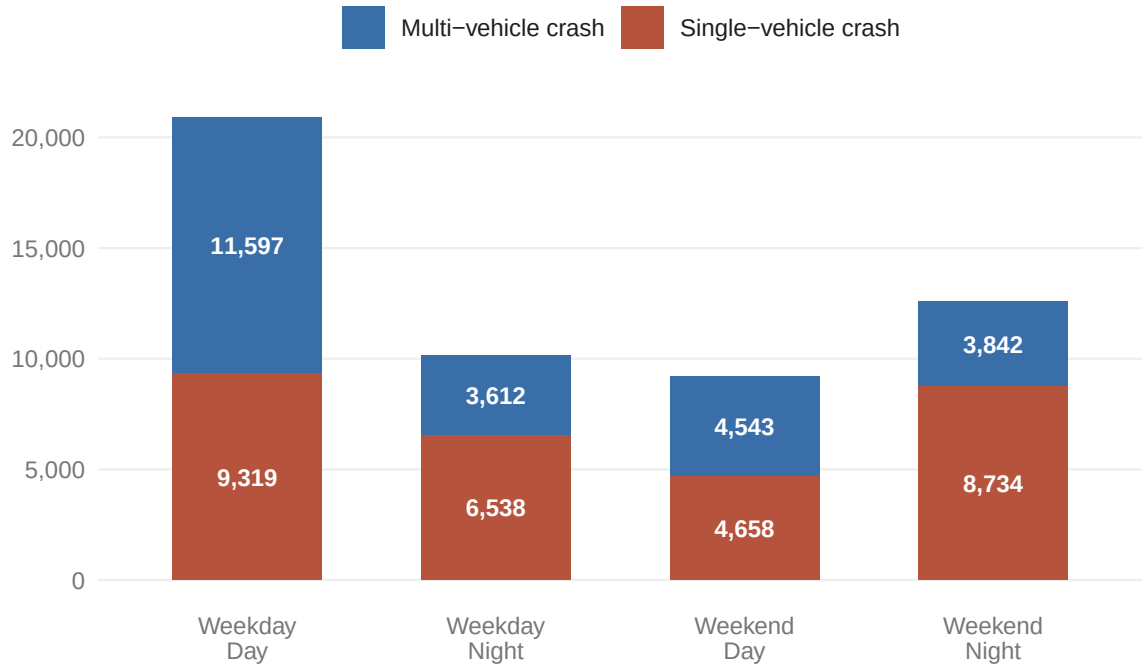


Figure 4: Stacked bar chart — fatalities by time slot and crash type. Numbers inside bars are counts.

Four time slots, 2 types of crashes. The theme seems to be evident. During weekday days most fatal crashes involve multiple vehicles. However, the picture turns at night – single vehicle crashes are predominant. A driver strays off the road, usually a country highway, usually by himself. The pattern accounts for 8,734 of the 12,576 deaths in that slot, which is 69 per cent of the deaths that occur on weekend nights.

Drivers bear the most

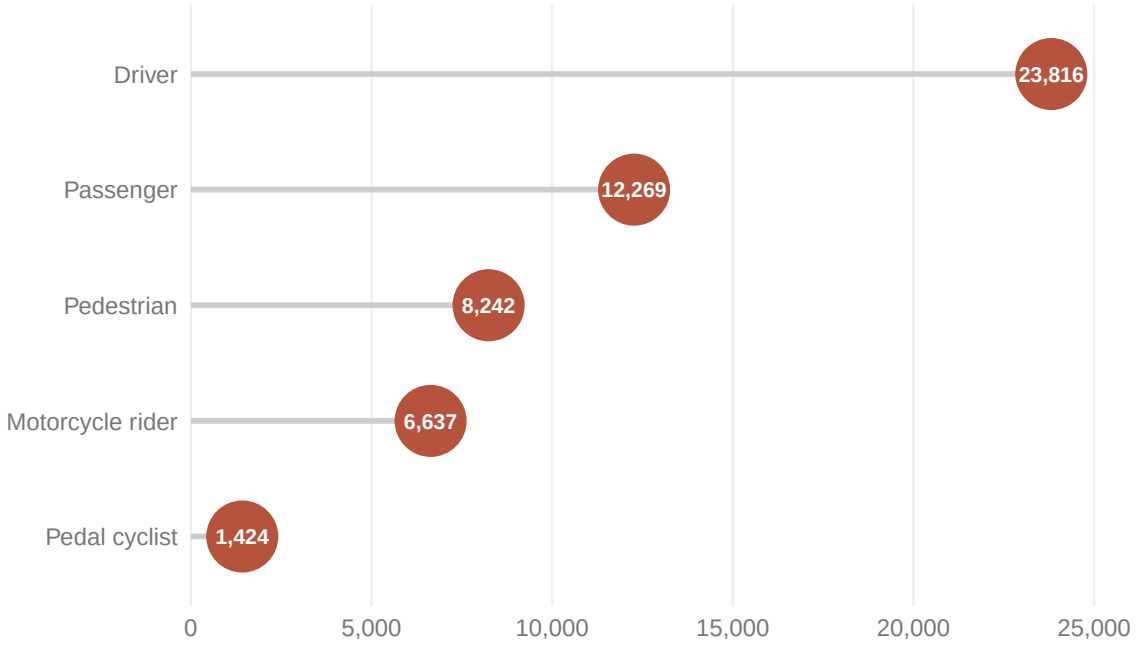


Figure 5: Lollipop chart — total fatalities by road user type, 1989–2021.

Drivers are involved in more fatalities than any other road user group. Passengers come in second — as a reminder that fatal crashes can have more than one life. Motorcyclists are also significantly over-represented: small percentage of all road transportation activity, and yet a consistently large percentage of all road transportation deaths.

The same young man, every chart

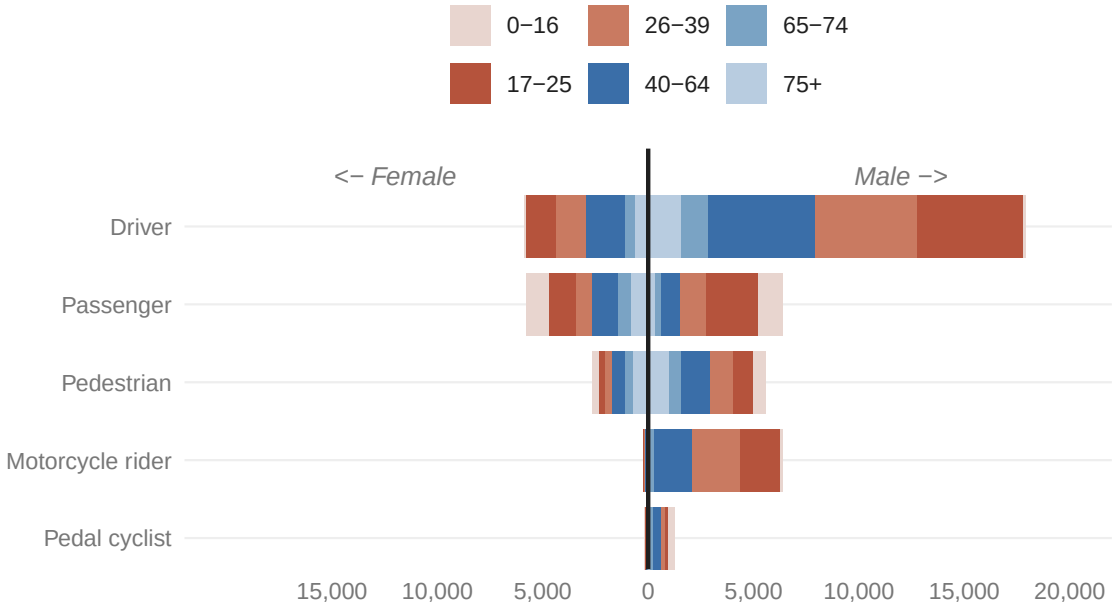


Figure 6: Diverging bar chart — fatalities by road user type, age group and gender. Female left, male right.

Male fatalities outnumber females in each road user group, by about 2.5 to 1. The longest single bar in the chart is the brick-red one for 17 to 25-year-old male drivers with 5,019 deaths (over a period of thirty-two years). Most people that are killed in pedestrian crashes are older. The death of motorcycle riders falls within the 26–39 age group.

What the six charts say together

The road safety improvement in Australia has been widespread, continued and substantial. It is displayed on the heatmap, one decade at a time. Year-by-year it is shown on the national trendline.

However, three charts contradict that sentiment. The slope chart indicates that WA and Queensland have improved the least. The stacked bar indicates that single-vehicle crashes continue to be a consistent problem on weekends at night, for all thirty-two years. As revealed in the diverging bar, young male drivers remain the largest single group of victims.

> *The story of Australian road fatalities is no longer one of national decline. A tale of unfulfilled dreams, late night, lonely country road.*

Data and methods

Data source

- Australian Road Deaths Database 1989–2021, Bureau of Infrastructure and Transport Research Economics (BITRE), accessed via Kaggle.
- 52,843 individual fatality records across 23 variables, covering one record per fatality.
- Variables used: year, month, state, time of day, day of week, crash type, road user type, age group, gender, and remoteness category.

Data cleaning

- Age column converted from character to numeric. Records with age outside 0–100 excluded for age-based figures only.
- One record with gender listed as "Unspecified" excluded from gender-based figures only.
- Remoteness category available only from 2014 onward and not used in trend charts; state is used as a geographic proxy for the full 1989–2021 range.
- All other records retained. Cleaning was selective by variable, not applied globally, to preserve sample size.

Visualisation types used

- Figure 1: Filled line chart
- Figure 2: Calendar heatmap (tile grid)
- Figure 3: Slope chart (connected dot plot)
- Figure 4: Stacked bar chart
- Figure 5: Lollipop chart
- Figure 6: Diverging horizontal bar chart

Chart type selection followed Cleveland and McGill's (1984) hierarchy of accurate encodings, with position on a common scale preferred for comparison. In-bar labels and minimal gridlines apply Tufte's (1983) data-ink principle. The Gestalt principle of proximity used in the heatmap layout draws on Bertin (1983). The overall narrative structure follows the approach outlined in Knaflic (2015) and Cairo (2016).

References

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Cleveland, W. S., & McGill, R. (1984). Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of the American Statistical Association*, *79*(387), 531–554. https://doi.org/10.2307/2288400

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Tufte, E. R. (1983). *The Visual Display of Quantitative Information*. Graphics Press.